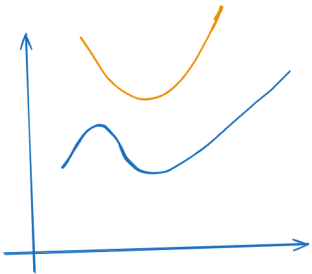
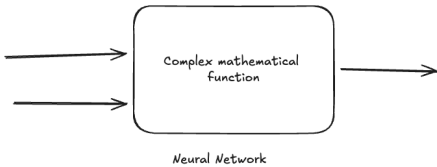
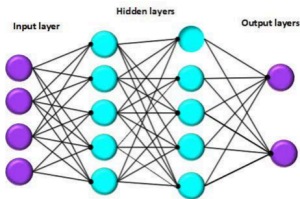
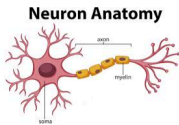
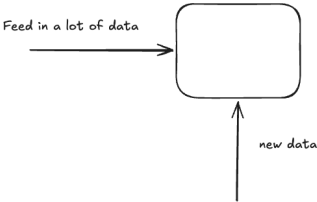
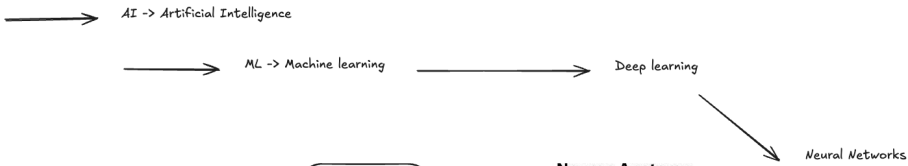


Introduction to LLMs

Large language model



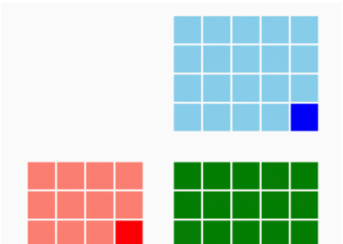
X -> input
Y -> output

Original data -> X = 10, Y = 30

Line 1 -> X = 10, Y = -9
Line 2 -> X = 10, Y = 28

MSE -> mean square error
Absolute error -> $|Y_{pred} - Y_{origin}|$

-> In major neural networks you will see one major type of operation:
-> Matrix multiplication



B -> Blue

GPU -> Graphic processing unit

A -> Red

Dimensions of A -> 3x4
Dimensions of B -> 4x5

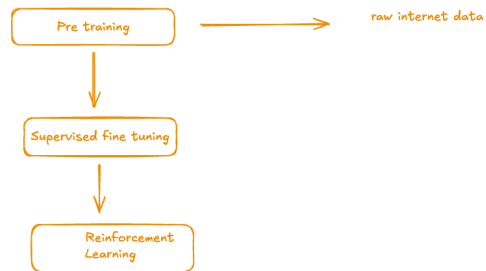
A x B -> MxN rule

$A_{11} \cdot B_{11} + A_{12} \cdot B_{21} + \dots$

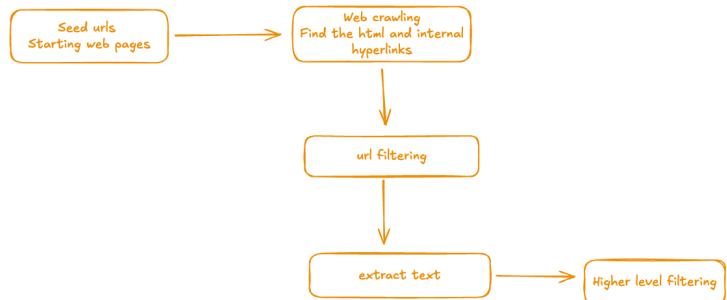
Transformers -> A word's importance or meaning is heavily influenced by it's neighbours, and the context in which we are speaking / saying it.

Attention -> this mechanism lets the network decide how the other words influence the current word under consideration.

How LLMs actually came up.



Web scraper



HuggingFaceFW / blogpost-fineweb-v1

FineWeb: decanting the web for the finest text data at scale

...

Similar datasets can be used for training a model



Tokenization -> Mechanism with which we convert raw text data into a sequence of integer IDs from a given vocabulary.

Tokens -> Tokens are the fundamental unit that neural networks can process.

LLMs are a great invention.

bits -> bytes -> Tokens

